



STATUS OF THE WORLD'S SOIL RESOURCES 2026

A global call to action to restore and sustain the world's soils

KEY MESSAGES

- Soil resources are **under mounting pressure** from a combination of socioeconomic and environmental challenges. Over half of all regions soil assessments have indicated as *poor* or *very poor* soil stewardship efforts, and current measures are insufficient to halt or reverse soil degradation in many places.
- Overall, soil management remains largely **unsustainable** worldwide. Soil erosion remains the most severe threat to soil health globally, although all threats are interconnected and often reinforce one another through cascading degradation processes that affect multiple soil functions simultaneously.
- A critical **implementation gap** persists between proven solutions and their adoption on the ground at the scale needed to avoid, reduce and reverse soil degradation. Roughly half of regions report low or very low adoption of sustainable soil management practices and only ~10 percent show improving trend since 2015.
- Closing the gap between knowledge and action demands **immediate, coordinated action**: enabling policies, incentives, capacity building and strong collaboration across stakeholders are essential to scale up sustainable soil management and safeguard soils for future generations.

Soil health: The foundation for productive, resilient and sustainable agrifood systems

Soils are the foundation of agrifood systems, producing over 95 percent of the food we eat and underpinning food security. Degraded soils yield less and are more prone to droughts and pests, directly threatening livelihoods and nutrition, especially in agrarian economies.

Yet, soils are not only the substrate for agricultural production but also living ecosystems that sustain rich biodiversity and provide a wide range of ecosystem functions and services. While soil formation takes centuries, degradation can occur rapidly under unsustainable management practices. Protecting soil health is therefore fundamental to ensuring the resilience, sustainability and long-term productivity of the agrifood sector.

Drivers, including population growth, rural-to-urban migration, economic development, political instability and rising inflation, shape land-use decisions, resource availability and governance conditions. These drivers are further exacerbated by climate change, which amplifies existing vulnerability. Since 2015, the global population has grown by 0.8 billion, driving higher demand for food, feed, fibre, fuel, and other commodities. This demand has led to a 24 percent increase in the harvested area of major crops between 2000 and 2022, including 80 million additional hectares since 2015. These land-use changes have caused a net loss of 4.7 million hectares of forest annually from 2010 to 2020, with significant regional

variation. At the same time, production intensification has driven higher use of agrochemicals.

Together, these drivers give rise to multiple pressures, including land abandonment, land-use change, unsustainable agricultural practices, industrial activities, conflicts and pollution, which collectively accelerate land degradation and the decline of soil functions.

Interlinked soil threats drive a degradation cascade, demanding integrated solutions

Soil degradation is a critical global problem that threatens food security, water resources, biodiversity and climate resilience. The *Status of the World's Soil Resources 2026* (SWSR 2026) reveals that soil management remains largely unsustainable worldwide. Despite increased research and awareness since 2015 on how better management can reduce soil threats, the reality on the ground is that the deterioration of soils continues at an alarming rate.

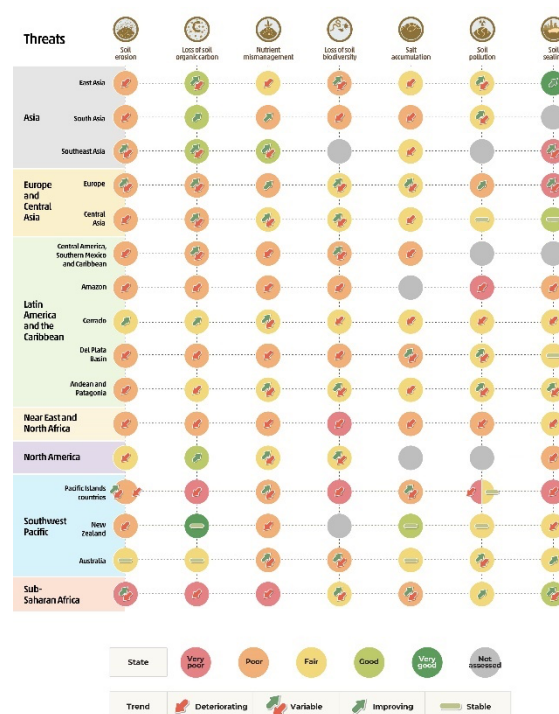
Overall, the global picture is sobering. **Soil erosion** remains the most severe threat to soils worldwide. It is also the most interconnected threat, stripping away topsoil, reducing rooting depth, depleting nutrients and carbon, limiting water infiltration, and increasing runoff, thereby heightening the risk of floods and droughts. The **loss of soil organic carbon (SOC)** is a major concern, especially where deforestation, peatland drainage or unsustainable practices have depleted organic matter. **Nutrient mismanagement**, whether through overuse of fertilizers leading to pollution or inadequate fertilizer resulting in nutrient mining of soils is similarly widespread. Particularly concerning is the continued **deterioration of soil biodiversity**, which has emerged as one of the most rapidly deteriorating soil threat globally.

Salinization (salt build-up in soils) challenges arid and semi-arid regions and low-lying coasts, while **soil pollution** (from industrial activities, mining, waste and agrochemicals) is a serious issue for many populated and agricultural areas. **Urban expansion and soil sealing** – the covering of fertile land with impermeable surfaces – continue to claim millions of hectares, particularly around fast-growing cities, undermining local food production and water cycle regulation. These threats are **not isolated**, as many landscapes experience multiple overlapping forms of soil degradation, further complicating restoration efforts.

The **effectiveness of management practices** adopted since 2015 in preventing, minimizing and reversing the majority of these soil threats is assessed as ranging from

fair to very poor (Figure 1). Across most regions, soil threats have either persisted or worsened over the past decade, indicating that current efforts have been insufficient to halt degradation or restore soil health at the scale and pace required. These trends point to limited progress in the widespread adoption and effective implementation of sustainable soil management (SSM) practices. Unless reversed, continued soil degradation will undermine global efforts to achieve land degradation neutrality and the Sustainable Development Goals (SDGs) (Figure 2).

Figure 1. Global assessment of state and trend in the adoption of sustainable soil management practices to address threats to soil functions across regions (Click to enlarge)



Note: Coloured circles depict the current state of adoption of SSM practices to address each threat in that region. The small arrows show the trend in SSM adoption since 2015.

Source: FAO and ITPS. 2026. *Status of the World's Soil Resources 2026 – Main report*. Rome. <https://doi.org/10.4060/ce0792en>

Scientific knowledge on sustainable soil management has expanded, but adoption remains far below the scale needed

The *Revised World Soil Charter*¹ and the *Voluntary Guidelines for Sustainable Soil Management (VGSSM)*² establish the principles for SSM. This approach encompasses a broad range of practices and is widely recognised as a cost-effective strategy for maintaining and restoring soil health. A substantial body of scientific

¹ FAO. 2015. *Revised World Soil Charter*. Rome. <https://openknowledge.fao.org/handle/20.500.14283/i4965e>

² FAO. 2016. *Voluntary Guidelines for Sustainable Soil Management*. Rome. <https://openknowledge.fao.org/handle/20.500.14283/bl813e>

evidence confirms that many SSM practices – including reduced tillage, integrated nutrient management, crop diversification, cover crops, organic amendments and agroforestry – can effectively mitigate soil threats while sustaining or improving agricultural productivity.

Long-term studies conducted across diverse geographic regions, soil types, climate conditions, and cropping systems consistently show that there is no one-size-fits-all solution. The effectiveness of individual practices depends on local biophysical and socioeconomic conditions, highlighting the need for context-specific management approaches.

Despite their proven benefits, SSM practices are not being adopted at the scale required to reverse global soil degradation. Evidence compiled in SWSR 2026 reveals a **major implementation gap**: in many regions, the uptake of SSM practices is too limited to prevent ongoing soil degradation. About **52 percent** of regional and subregional assessments **report low or very low levels of SSM adoption**, while 58 percent of assessments reported deteriorating trends since 2015, as illustrated in Figure 1.

Adoption is often constrained by limited access to financial, technical and institutional resources rather than lack of awareness. High upfront investment costs, together with the delayed realization of many soil-related benefits, often discourage farmers from adopting SSM practices.

Scaling up the adoption of SSM can be supported by comprehensive cost-benefit assessments that capture not only short-term productivity gains but also long-term environmental, economic and ecosystem service benefits, providing a stronger basis for policy design and public and private investment.

Bridging the implementation gap: Enabling conditions for scaling up action

There are continuing **systemic barriers** facing the widespread adoption of SSM practices, including inadequate soil information, limited extension services and training, insufficient integration of local and Indigenous knowledge, gender-based constraints and fragmented governance structures. Collectively, these factors weaken the enabling environment needed to scale up SSM.

Some SSM practices provide limited short-term gains in crop yields or require substantial upfront investments in technologies or inputs. In such cases, **targeted incentives** are critical to overcoming financial and behavioural barriers to adoption.

However, the effectiveness of these incentives depends on their economic viability. When payments or other benefits fail to offset the direct, indirect and opportunity costs associated with adopting new practices, farmers are unlikely to change their management. To be effective, incentive frameworks must be well-targeted, context-specific and integrated into broader policies that support sustainable land and soil management. They should also be complemented by strong institutions, robust monitoring systems and inclusive dialogue with stakeholders at all levels.

Significant progress has been made in improving the availability and quality of soil data at both national and international levels, increasingly guided by the FAIR principles (Findable, Accessible, Interoperable and Reusable). However, this growing body of information has not yet fully translated into **decision-support tools** tailored to the diverse needs of policymakers, extension services and land managers. National-scale soil data often generate generalized recommendations that require skilled local advisors to adapt them to site-specific conditions. Where **local advisory capacity** is lacking, recommendations risk being ineffective or even counterproductive. The challenge of scale further limits practical application. While regional and global models provide useful overviews, they often fail to reflect ground-level realities due to insufficient ground validation. Farm-level decision-making requires detailed soil information at fine spatial resolutions (e.g. 1:5 000 scale), whereas many existing maps remain available at much coarser resolutions (e.g., 1:100 000 or 1:250 000).

Integrating local, traditional and Indigenous knowledge is an essential but frequently overlooked component of an effective enabling environment for SSM. Respecting and applying these knowledge systems requires sustained dialogue with communities, the co-development of solutions, and the joint evaluation of knowledge across scientific and local perspectives. Initiatives that neglect this risk low acceptance and limited long-term success.

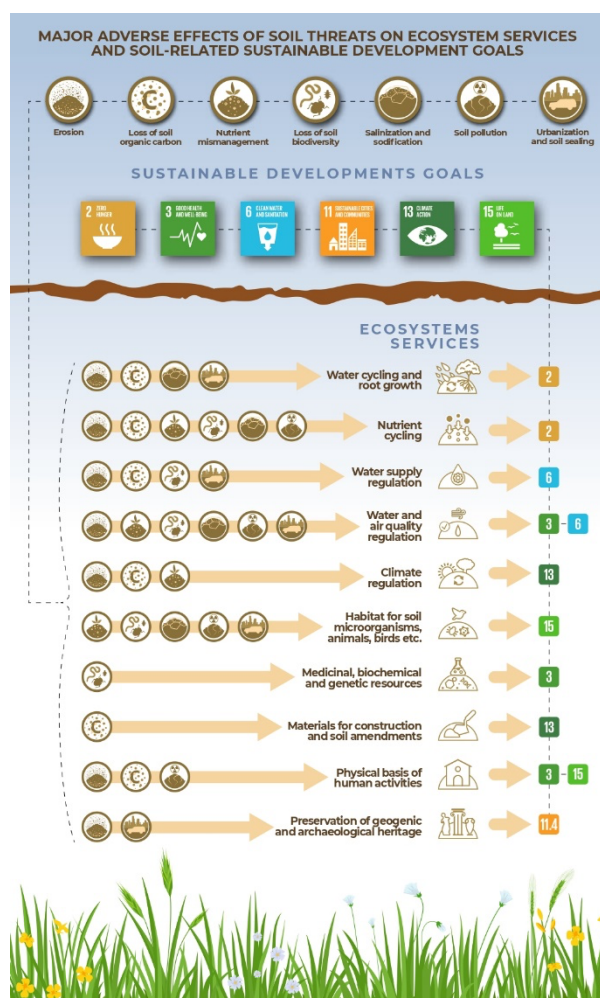
National policies and governance frameworks must align with local decision-making structures. Overly prescriptive, top-down approaches often fail to account for the complexity of local institutions and socioeconomic contexts, creating barriers that undermine implementation. National strategies that provide a coherent policy framework while allowing sufficient flexibility for local adaptation are essential to ensure effective and sustainable adoption of SSM practices.

Scaling up SSM globally will require integrated interventions that combine enabling policies, well-

designed incentive schemes, knowledge generation and dissemination, strengthened advisory services, and improved access to financial and technical resources, all tailored to regional contexts and local constraints. Establishing these enabling conditions will enable countries to accelerate the adoption of SSM practices and scale successful approaches across landscapes and regions.

Figure 2. Major adverse effects of soil threats on ecosystem services and soil-related Sustainable Development Goals as identified in the Status of the World's Soil Resources report 2026

(Click to enlarge)



Source: FAO and ITPS. 2026. *Status of the World's Soil Resources 2026 – Main report*. Rome. <https://doi.org/10.4060/ce0792en>

Call to Action: Safeguarding soil health for our future

Despite the increasing complexity and urgency of long-standing and emerging socioeconomic and environmental challenges, the core solutions to addressing soil threats remain clear. The principles set out in the **Revised World Soil Charter** are as relevant today as when first endorsed. However, their implementation remains uneven and insufficient across countries and sectors.

As SWSR 2026 demonstrates, reversing global soil degradation is an **urgent priority** requiring **coordinated and sustained action** by governments, the private sector, researchers, farmers and civil society. High-level decision-makers are urged to recognize healthy soils as **strategic assets** that underpin food security, water security, climate change mitigation and adaptation, biodiversity conservation, and resilient livelihoods. **Investing in soil health yields substantial socioeconomic and environmental returns**, far outweighing the costs of inaction.

Action is needed at local, national, regional and international levels, guided by five priorities:

1. strengthening soil governance and legal frameworks;
2. promoting education, capacity development and knowledge exchange;
3. improving soil monitoring through harmonized soil data, indicators and assessment methods;
4. enhancing incentive frameworks to accelerate the adoption of SSM; and
5. fostering inclusive, multidisciplinary and multistakeholder decision-making.

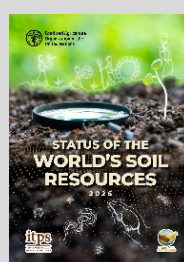
Globally, we must shift from a mindset of soil exploitation to one of soil stewardship. The knowledge, tools and practices needed to protect and restore soils already exist. The challenge is no longer knowing what needs to be done, but acting with the urgency, ambition, long-term commitment and at the scale and pace required to safeguard soils for current and future generations.

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